Lecture 4

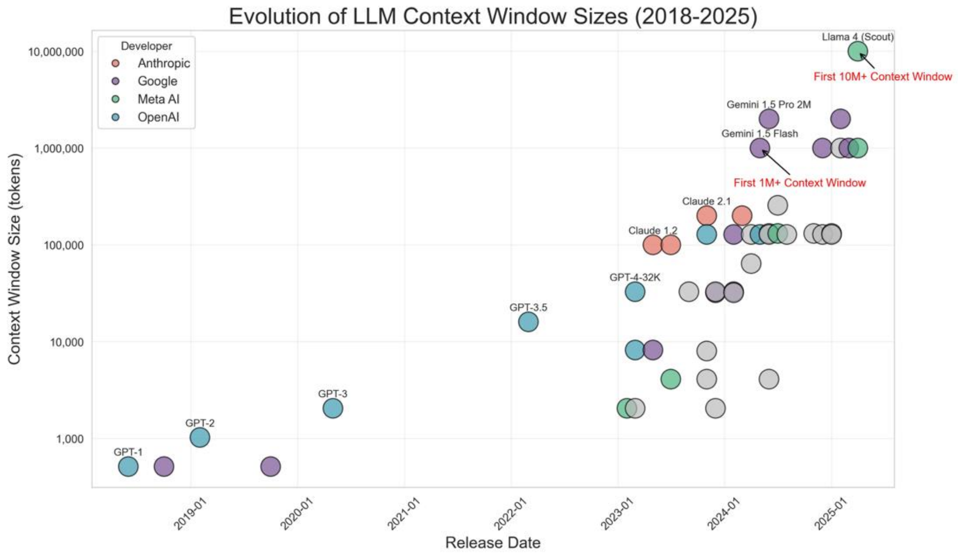
Attention Alternatives and Mixtures of Experts

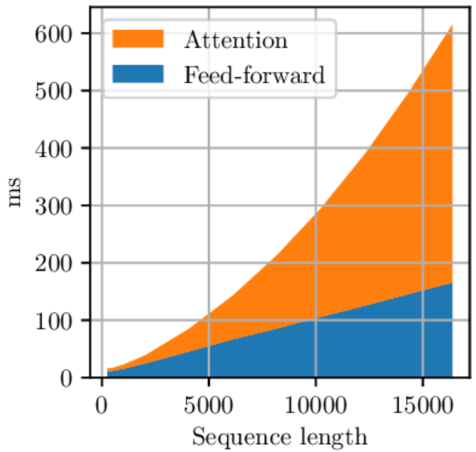
CS336

Tatsu H

Attention alternatives

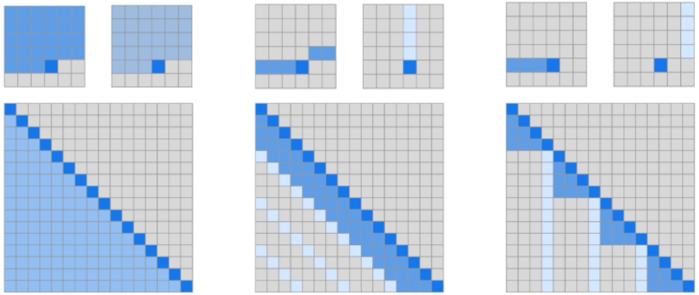
Cost of attention rises with large context sizes… how do we control those costs?



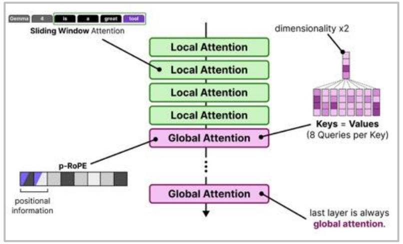


https://www.meibel.ai/post/understanding-the-impact-of-increasing-llm-context-windows

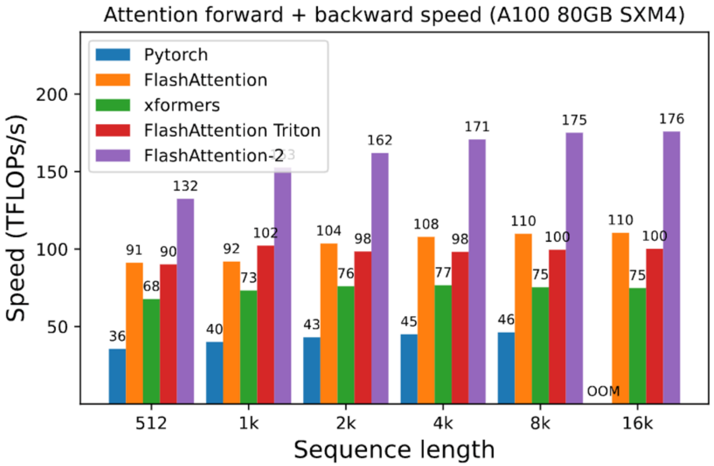
The ‘basic’ toolkit



(a) Transformer (b) Sparse Transformer (strided) (c) Sparse Transformer (fixed)



Combine local + global attention



Systems engineering

But what if we want more radical and potentially large gains?

Linear attention

Consider the usual attention operation.. $Q \in R^{n \times d_k}, K \in R^{n \times d_k}, V \in R^{n \times d_v}$

$$Attn(Q, K, V) = \rho(QK^\top)V$$

This is quadratic due to $QK^\top$ $(n^2 d_k)$. Can we do better (when ρ is the identity)?

$$(QK^\top)V = Q(K^\top V)$$

This is very silly, but surprisingly important.. We get from $n^2 d_k + n^2 d_v$ to $2nd_v d_k$

[Shen et al 2018, Katharopoulos 2020 for kernel version. Also connections to things like fast weight programmers]

Recurrent form of linear attention

Recall that in purely linear attention, we consider the reordering

$$(QK^\top)V = Q(K^\top V)$$

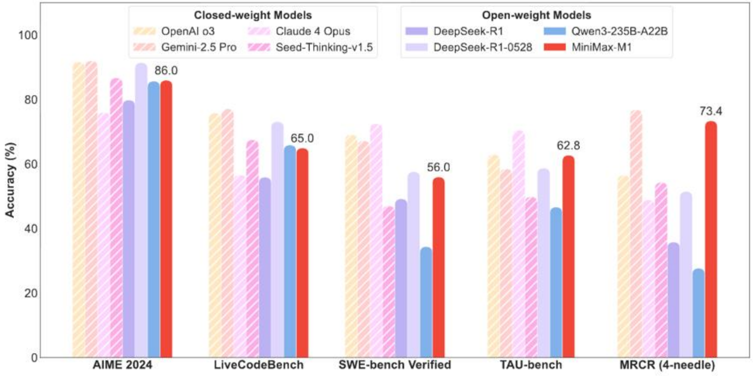
This is linear time (great) but the even nicer thing is that this looks like a RNN.

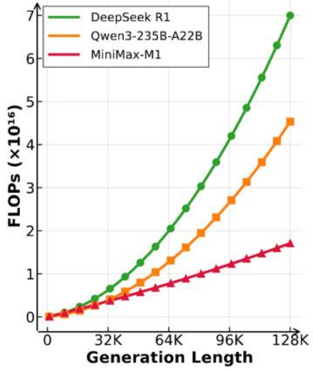
$$S_t = S_{t-1} + k_t v_t^\top \quad \text{and} \quad y_t = q_t^\top S_t$$

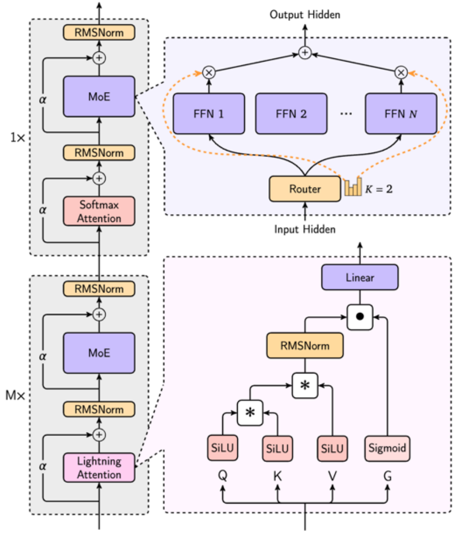
This ‘duality’ allows us to train efficiently (using the parallel, quadratic form) and inference efficiently (using the serial, linear form)

(Note: if one weights S_{t-1} by γ, you get a RetNet)

Minimax M1







Arch.	BBH ↑	DROP ↑	MMLU ↑	CMMLU ↑	MATH ↑	GSM8k ↑	ARC-C ↑	WG ↑
Softmax	28.2	27.4	49.3	**47.3**	4.6	**18.8**	46.4	65.6
Hybrid-lightning	**32.2**	**29.0**	**49.5**	46.0	**6.8**	18.5	**47.4**	**67.5**

Minimax M1 (and minimax-text-01) use a 7-to-1 hybrid (7 linear, 1 full) linear attention.

Performance (generally) seems strong, and they show linear scaling in context length

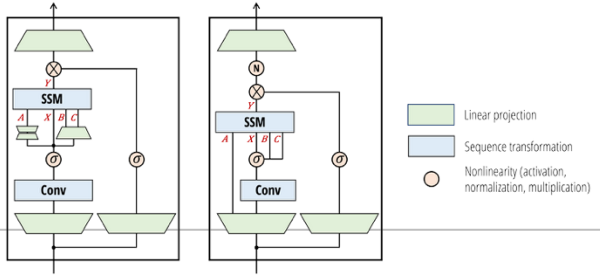
From linear attention to Mamba-2

Let's generalize linear attention a little bit and add per-position weights..

$$S_t = S_{t-1} + k_t v_t^\top \quad \text{and} \quad y_t = q_t^\top S_t \text{ (Linear attention)}$$
$$S_t = \gamma_t S_{t-1} + k_t v_t^\top \quad \text{and} \quad y_t = q_t^\top S_t + v_t^\top D \;\text{ with } \gamma_t = f(x_t) \text{(Mamba-2)}$$

There is a lot more words to justify this (go read the mamba 2 paper) but the mechanics is that we can make linear attention more expressive via gating (gating is good!)

This continues to have duality properties (compute γ in parallel, apply duality)



Sequential Mamba Block   Parallel Mamba Block

Nemotron 3

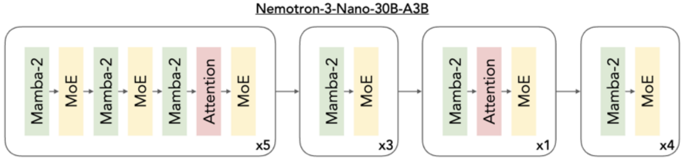


Figure 1 | Nemotron 3 models (e.g., Nemotron Nano 3) leverage a hybrid Mamba-Transformer MoE architecture consisting predominantly of interleaved Mamba-2 and MoE layers, with a select few self attention layers.

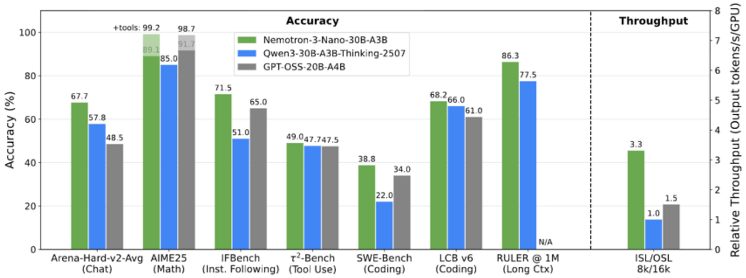


Figure 2 | The hybrid Mamba-Transformer MoE architecture used by Nemotron 3 models can achieve state-of-the-art accuracy on leading reasoning benchmarks and ultra-long-context tasks while providing throughput improvements over similarly sized Transformer MoEs. For details, please see the Nemotron Nano 3 technical report.

Mamba attention hybrid (3-1 ish) – comparable (or better) pref to other similar models

Gated delta net (and friends)

Let’s generalize things further – gate the input and selectively erase the state.

$$S_t = \gamma_t S_{t-1} + k_t v_t^\top \quad \text{and} \quad y_t = q_t^\top S_t + v_t^\top D \ \text{ with } \gamma_t = f(x_t) \text{(Mamba-2)}$$

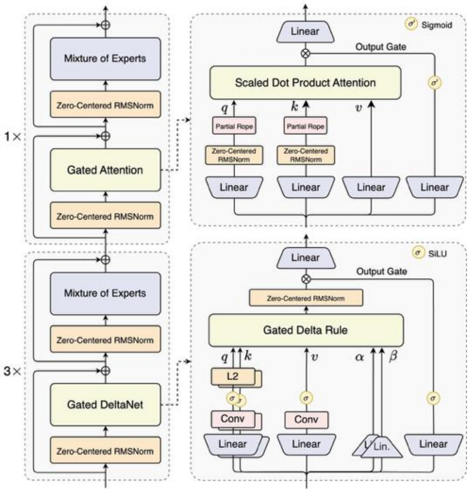
$$S_t = \gamma_t (I - \beta_t k_t k_t^\top) S_{t-1} + \beta_t k_t v_t^\top \quad \text{and}$$
$$y_t = q_t^\top S_t \text{ with } \gamma_t = f(x_t), \beta_t = f(x_t) \text{ (Gated Delta Net)}$$

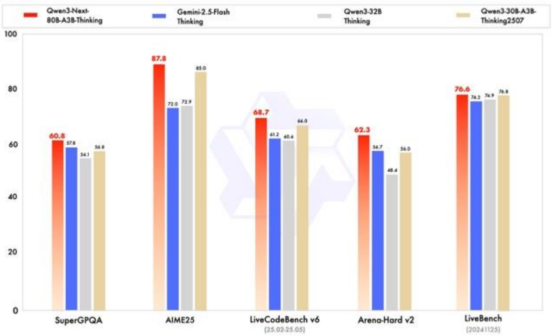
The gated delta net adds a ‘no input operation’ gate ($\beta = 0$)

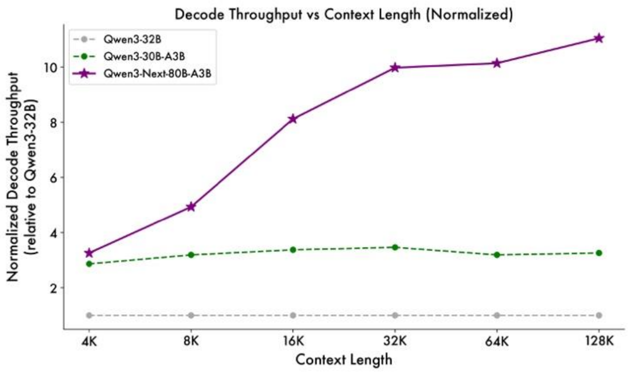
And *erases* anything in the direction of the current key $(I - \beta_t k_t k_t^\top)$

Close relationships to various fast weight programming / test time training ideas.

Qwen 3.5 / Qwen Next







The newest qwen are 3-1 GDN / Attention hybrids.

Once again, pretty reasonable performance with good inference characteristics

Hybrid performance

ByteDance | Seed

UC SANTA CRUZ

A Systematic Analysis of Hybrid Linear Attention

Dustin Wang*[1], Rui-Jie Zhu*[1,2], Steven Abreu[3], Yong Shan[2], Taylor Kergan[1], Yuqi Pan[2,4], Yuhong Chou[5], Zheng Li[2], Ge Zhang[2,6,†], Wenhao Huang[2], Jason Eshraghian[1,†]

[1]UC Santa Cruz, [2]ByteDance Seed, [3]University of Groningen, [4]CASIA, [5]PolyU, [6]M-A-P

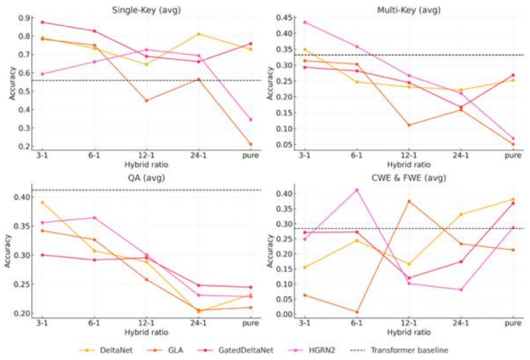
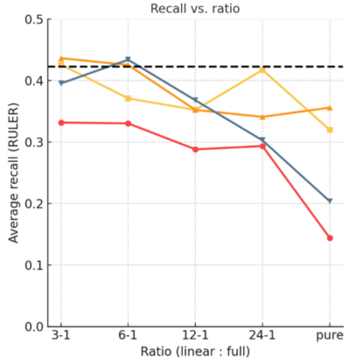


Figure 4 RULER sub task results based on ratio. RetNet and HGRN model families are omitted as their recall benchmark results were insignificant.

Model	Update rule (S_t)	Read-out (o_t)
Vector-valued hidden state (classical / gated RNNs)		
HGRN[4]	$h_t = \alpha_t \odot h_{t-1} + (1-\alpha_t) \odot v_t$	$o_t = h_t \odot q_t$
Hawk (RG-LRU)[33]	$h_t = r_t\, h_{t-1} + i_t \odot x_t$	$o_t = h_t \odot q_t$
Matrix-valued state via outer products		
RetNet/Lightning [11, 17]	$S_t = \gamma S_{t-1} + v_t k_t^\top$	$o_t = S_t q_t$
GLA[34]	$S_t = S_{t-1} \odot (1\alpha_t^\top) + v_t k_t^\top$	$o_t = S_t q_t$
Mamba-2[35]	$S_t = \gamma_t\, S_{t-1} + v_t k_t^\top$	$o_t = S_t q_t$
RWKV-6[36]	$S_t = S_{t-1}\mathrm{Diag}(\alpha_t) + v_t k_t^\top$	$o_t = (S_{t-1} + (d \odot v_t) k_t^\top) q_t$
HGRN-2/MetaLA[4, 37]	$S_t = S_{t-1}\mathrm{Diag}(\alpha_t) + v_t (1-\alpha_t)^\top$	$o_t = S_t q_t$
Delta-rule / controlled-forgetting family		
DeltaNet[5]	$S_t = S_{t-1}(I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top$	$o_t = S_t q_t$
Gated DeltaNet[19]	$S_t = \alpha_t\, S_{t-1}(I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top$	$o_t = S_t q_t$



Not many controlled ablations, but some evidence of low losses at small hybrid ratios

Alternative to hybrids: sparse adaptation

Instead of attending to every token, do sparse attention (DSA)

Prototype of DSA. The prototype of DSA primarily consists of two components: a lightning indexer and a fine-grained token selection mechanism.

The **lightning indexer** computes the index score $I_{t,s}$ between the query token $\mathbf{h}_t \in \mathbb{R}^d$ and a preceding token $\mathbf{h}_s \in \mathbb{R}^d$, determining which tokens to be selected by the query token:

$$I_{t,s} = \sum_{j=1}^{H^I} w_{t,j}^I \cdot \mathrm{ReLU}\left(\mathbf{q}_{t,j}^I \cdot \mathbf{k}_s^I\right), \tag{1}$$

where H^I denotes the number of indexer heads; $\mathbf{q}_{t,j}^I \in \mathbb{R}^{d^I}$ and $w_{t,j}^I \in \mathbb{R}$ are derived from the query token $\mathbf{h}_t$; and $\mathbf{k}_s^I \in \mathbb{R}^{d^I}$ is derived from the preceding token $\mathbf{h}_s$. We choose ReLU as the activation function for throughput consideration. Given that the lightning indexer has a small number of heads and can be implemented in FP8, its computational efficiency is remarkable.

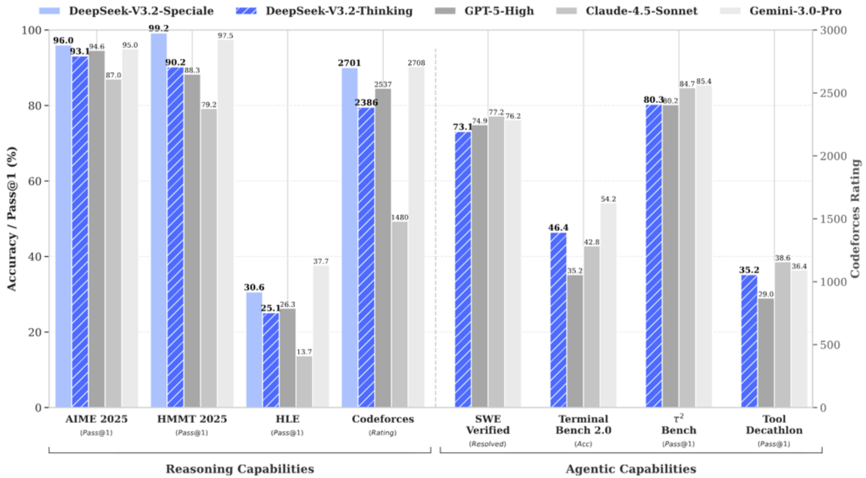
Given the index scores $\{I_{t,s}\}$ for each query token $\mathbf{h}_t$, our **fine-grained token selection mechanism** retrieves only the key-value entries $\{\mathbf{c}_s\}$ corresponding to the top-k index scores. Then, the attention output $\mathbf{u}_t$ is computed by applying the attention mechanism between the query token $\mathbf{h}_t$ and the sparsely selected key-value entries $\{\mathbf{c}_s\}$:

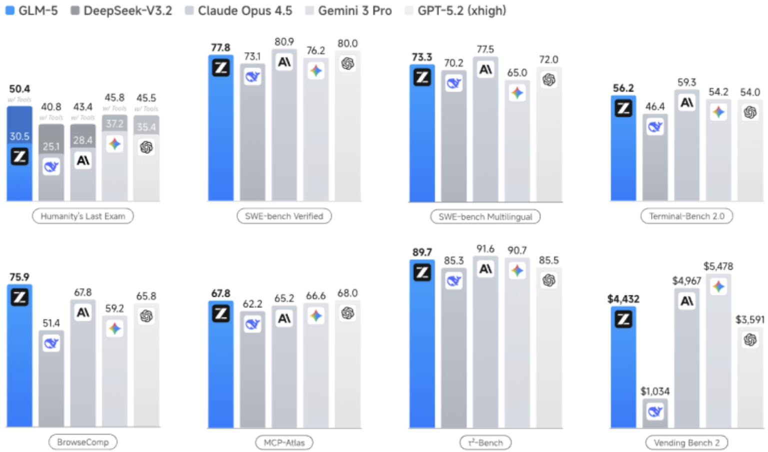
$$\mathbf{u}_t = \mathrm{Attn}\left(\mathbf{h}_t, \left\{\mathbf{c}_s \,\middle|\, I_{t,s} \in \mathrm{Top\text{-}k}(I_{t,:})\right\}\right). \tag{2}$$

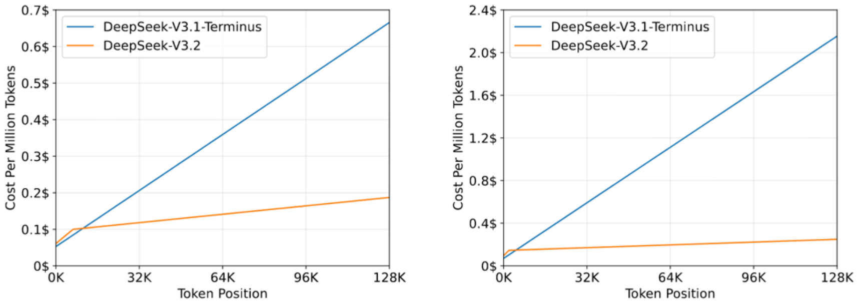
The indexer can be very lightweight, giving significant gains

Can be ‘post hoc’ adapted after dense short context pretraining

DSA – Deepseek Sparse Attention (v3.2, GLM5)





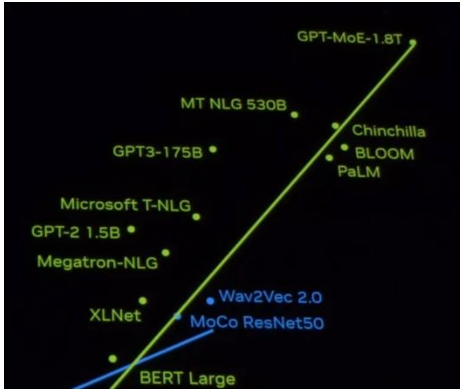


(a) Prefilling (b) Decoding

Table 6: RULER benchmark results for the GLM-4.7-Flash with DSA. The warmup-only variant trains only the indexer while keeping the base model frozen, the full DSA variant jointly trains both for 150B tokens.

	4K	8K	16K	32K	64K	128K
GLM-4.7-Flash	97.44	96.72	95.83	92.96	85.34	79.21
GLM-4.7-Flash + DSA warmup	97.51	96.54	95.40	90.09	84.05	71.35
GLM-4.7-Flash + DSA	96.77	96.25	96.69	93.45	87.06	78.86

Mixture of experts



GPT4 (?)

Mistral AI
@MistralAI

magnet:?xt=urn:btih:9238b09245d0d8cd915be09927769d5f7584c1c9&dn=mixtral-8x22b&tr=udp%3A%2F%2Fopen.demonii.com%3A1337%2Fannounce&tr=http%3A%2F%2Ftracker.opentrackr.org%3A1337%2Fannounce

Grok

deepseek

DeepSeekMoE: Towards Ultimate Expert Specialization in Mixture-of-Experts Language Models

deepseek

DeepSeek-V3 Technical Report

OLMoE: Open Mixture-of-Experts Language Models

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Sewon Min[a] Weijia Shi[w] Pete Walsh[a] Oyvind Tafjord[a] Nathan Lambert[a]
Yuling Gu[a] Shane Arora[a] Akshita Bhagia[a] Dustin Schwenk[a] David Wadden[a]
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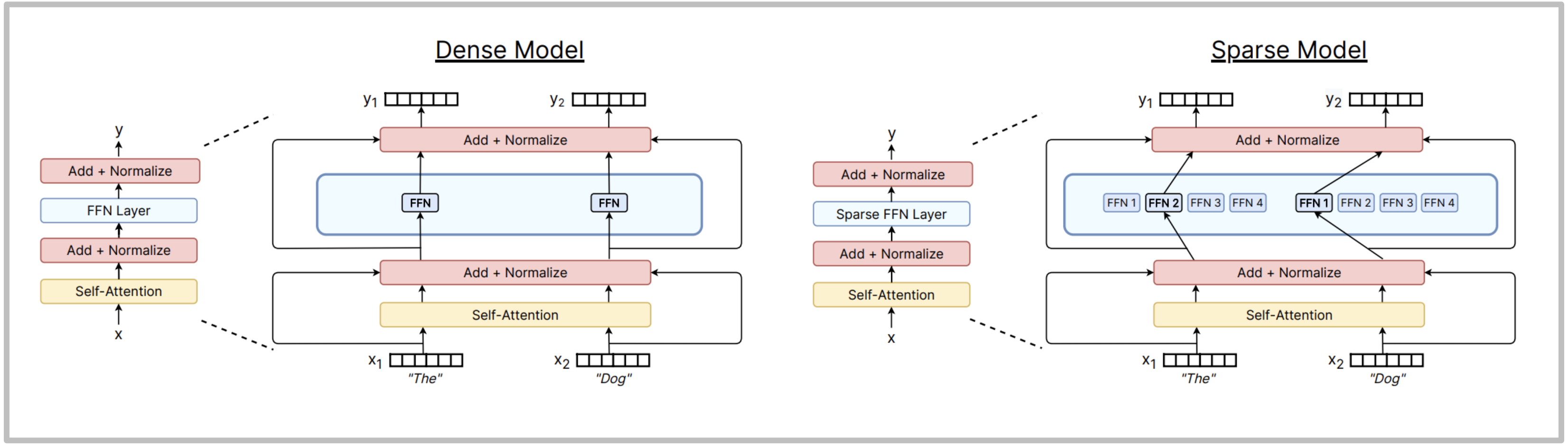
What's a MoE?



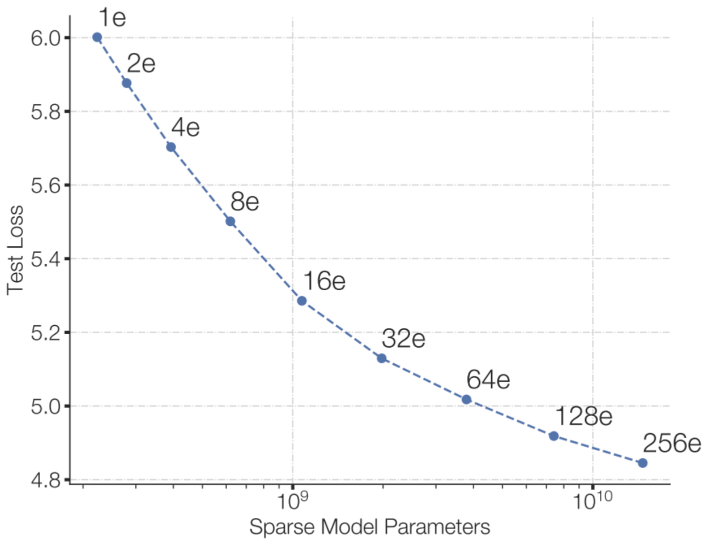
[Fedus et al 2022]

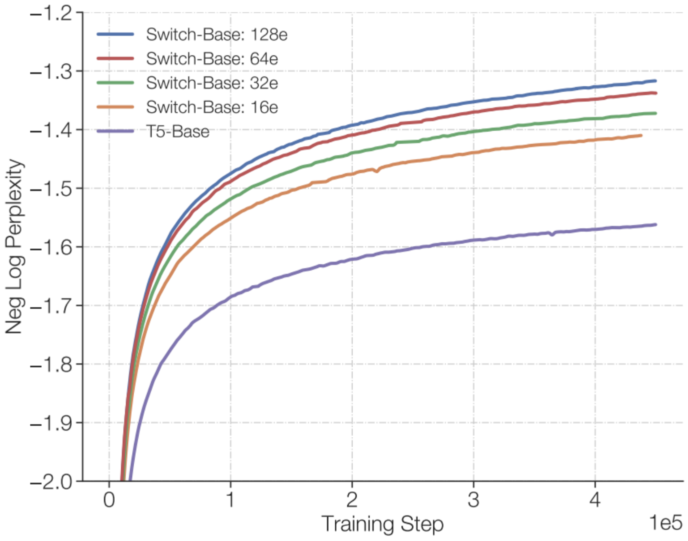
Replace big feedforward with (many) big feedforward networks and a selector layer

You can increase the # experts without affecting FLOPs

Why are MoEs getting popular?

Same FLOP, more param does better

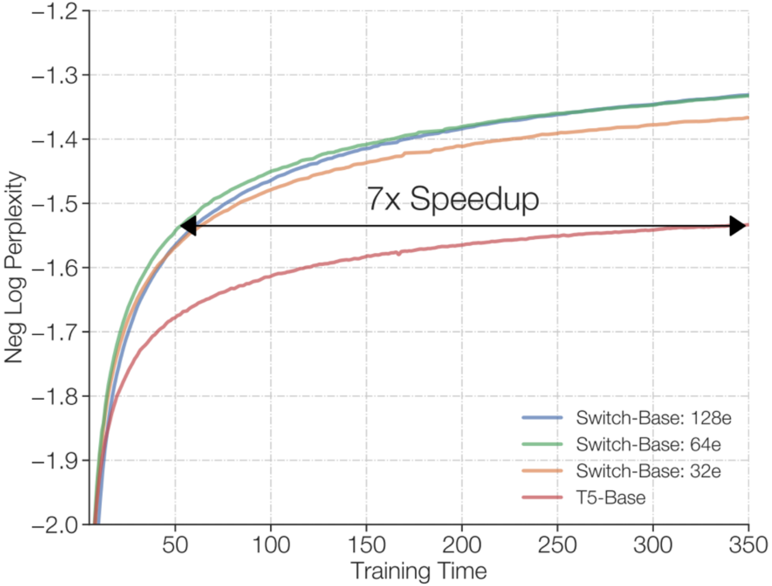




[Fedus et al 2022]

Why are MoEs getting popular?

Faster to train MoEs



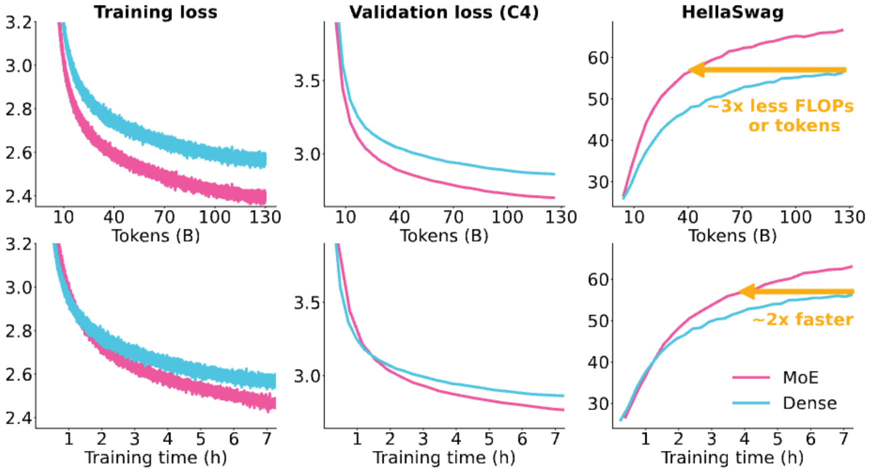
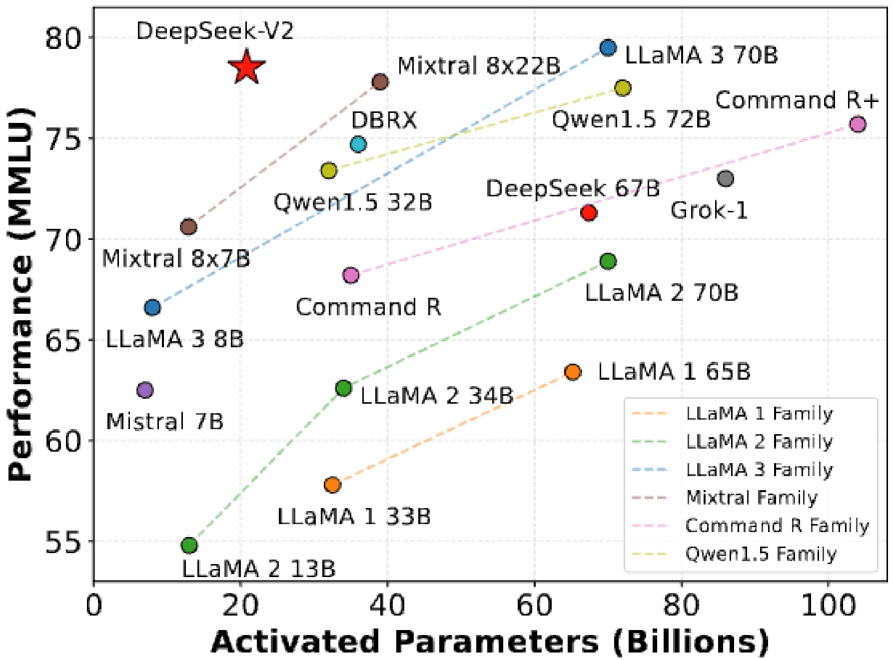


Figure 4: **MoE vs. Dense.** We train a 1.3B parameter dense model and a 1.3B active, 6.9B total parameter MoE model, each on 128 H100 GPUs. Apart from MoE-related changes, we train both

[OlMoE]

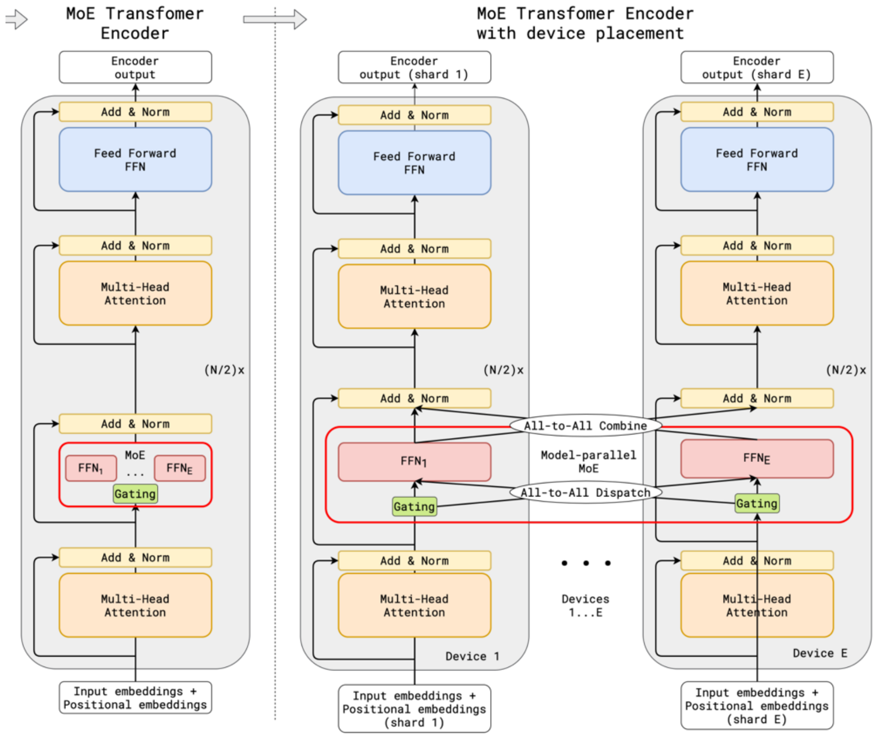
Why are MoEs getting popular?



Highly competitive vs dense equivalents

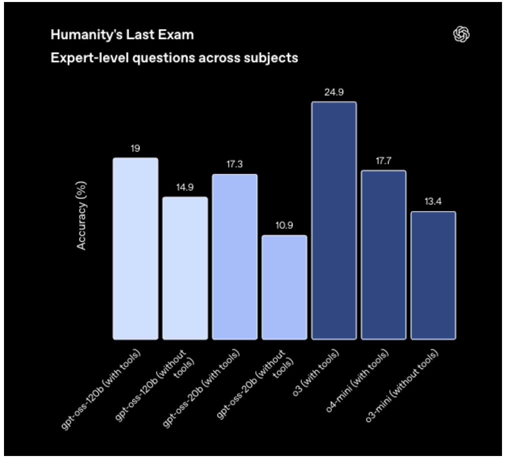
Why are MoEs getting popular?

Parallelizable to many devices



Some MoE results – from the west

Category / Benchmark	Llama 4 Maverick	Gemini 2.0 Flash	DeepSeek v3.1	GPT-4o
Inference Cost: Cost per 1M input & output tokens (3:1 blended)	$0.19-$0.49[5]	$0.17	$0.48	$4.38
Image Reasoning: MMMU	73.4	71.7	No multimodal support	69.1
MathVista	73.7	73.1		63.8
Image Understanding: ChartQA	90.0	88.3		85.7
DocVQA (test)	94.4	—		92.8
Coding: LiveCodeBench (10/01/2024-02/01/2025)	43.4	34.5	45.8/49.2[3]	32.3[3]
Reasoning & Knowledge: MMLU Pro	80.5	77.6	81.2	—
GPQA Diamond	69.8	60.1	68.4	53.6



MoEs are most of the highest-performance open models, and are quite quick

Earlier MoE results from Chinese groups – Qwen

Chinese LLM companies are also doing quite a bit of MoE work on the smaller end

Model	MMLU	GSM8K	HumanEval	Multilingual	MT-Bench
Mistral-7B	64.1	47.5	27.4	40.0	7.60
Gemma-7B	64.6	50.9	32.3	-	-
Qwen1.5-7B	61.0	62.5	36.0	45.2	7.60
DeepSeekMoE 16B	45.0	18.8	26.8	-	6.93
Qwen1.5-MoE-A2.7B	62.5	61.5	34.2	40.8	7.17

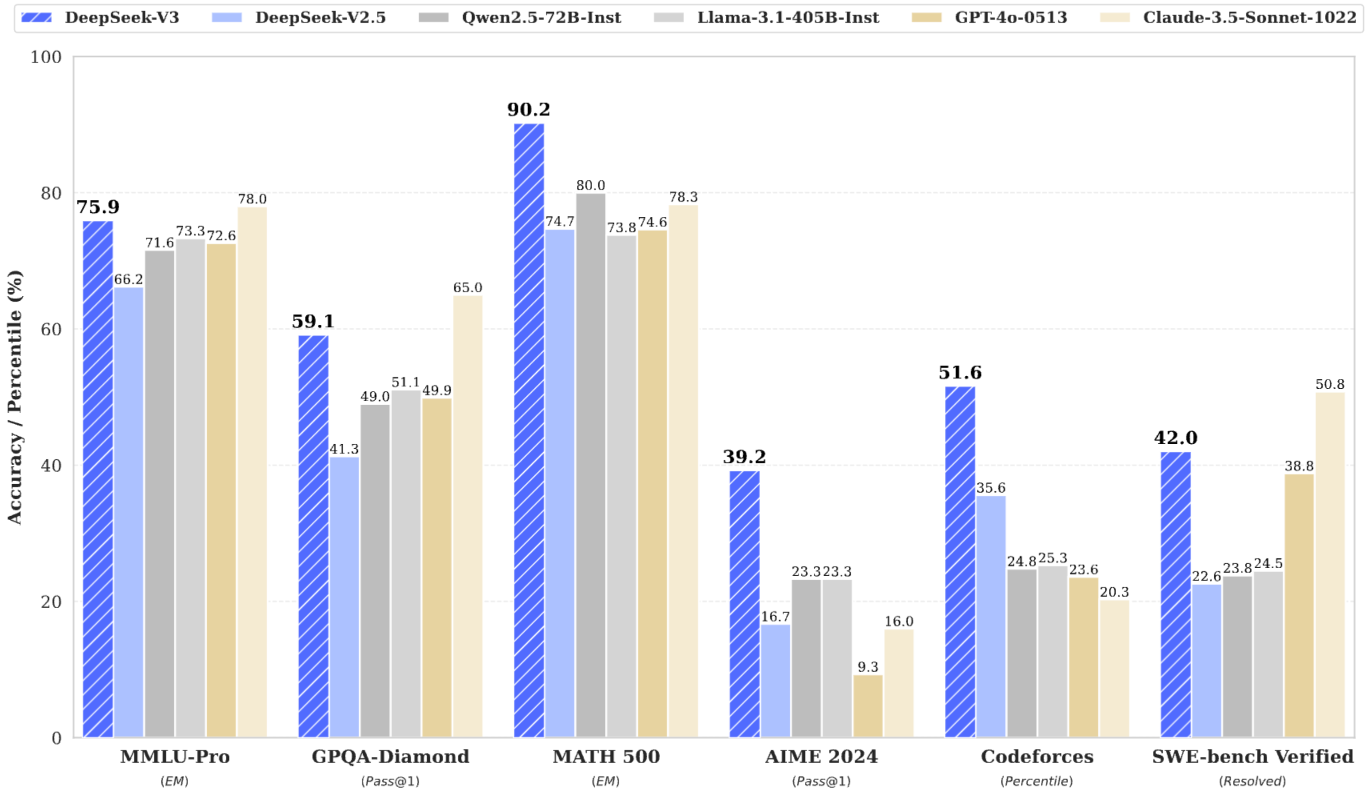
Model	#Parameters	#(Activated) Parameters
Mistral-7B	7.2	7.2
Qwen1.5-7B	7.7	7.7
Gemma-7B	8.5	7.8
DeepSeekMoE 16B	16.4	2.8
Qwen1.5-MoE-A2.7B	14.3	2.7

Earlier MoE results from Chinese groups - DeepSeek

There's also some good recent ablation work on MoEs showing they're generally good

Metric	# Shot	Dense	Hash Layer	Switch
# Total Params	N/A	0.2B	2.0B	2.0B
# Activated Params	N/A	0.2B	0.2B	0.2B
FLOPs per 2K Tokens	N/A	2.9T	2.9T	2.9T
# Training Tokens	N/A	100B	100B	100B
Pile (Loss)	N/A	2.060	1.932	1.881
HellaSwag (Acc.)	0-shot	38.8	46.2	49.1
PIQA (Acc.)	0-shot	66.8	68.4	70.5
ARC-easy (Acc.)	0-shot	41.0	45.3	45.9
ARC-challenge (Acc.)	0-shot	26.0	28.2	30.2
RACE-middle (Acc.)	5-shot	38.8	38.8	43.6
RACE-high (Acc.)	5-shot	29.0	30.0	30.9
HumanEval (Pass@1)	0-shot	0.0	1.2	2.4
MBPP (Pass@1)	3-shot	0.2	0.6	0.4
TriviaQA (EM)	5-shot	4.9	6.5	8.9
NaturalQuestions (EM)	5-shot	1.4	1.4	2.5

Recent MoE results – DeepSeek v3



Why haven't MoEs been more popular?

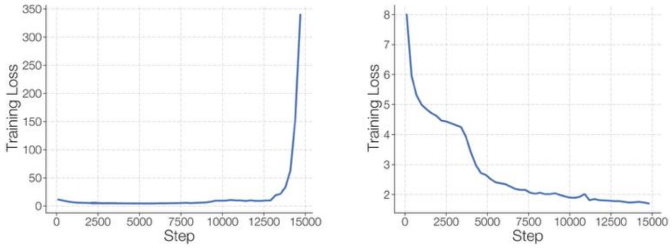
Infrastructure is complex / advantages on multi node

> At a high level, sparsity is good when you have many accelerators (e.g. GPU/TPU) to host all the additional parameters that comes when using sparsity. Typically models are trained using data-parallelism where different machines will get different slices of the training/inference data. The machines used for operating on the different slices of data can now be used to host many more model parameters. Therefore, sparse models are good when training with data parallelism and/or have high throughput while serving: training/serving on many machines which can host all of the parameters.

[Fedus et al 2022]

Training objectives are somewhat heuristic (and sometimes unstable)

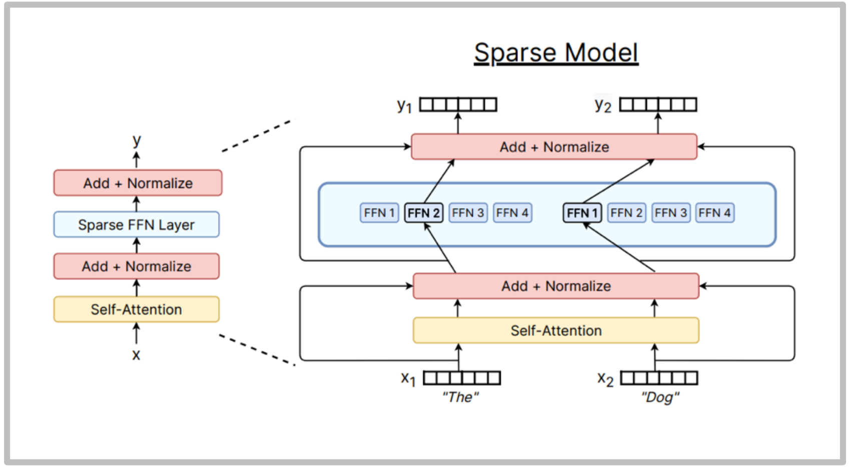
> Sparse models often suffer from training instabilities (Figure 1) worse than those observed in standard densely-activated Transformers.



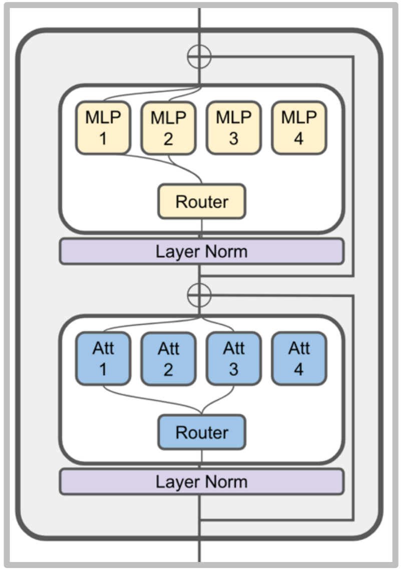
[Zoph et al 2022]

What MoEs generally look like

Typical: replace MLP with MoE layer



Less common: MoE for attention heads



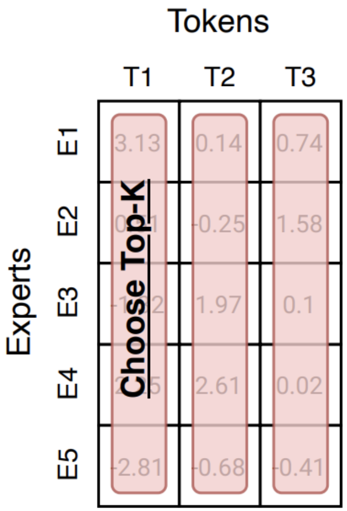
[ModuleFormer, JetMoE]

MoE – what varies?

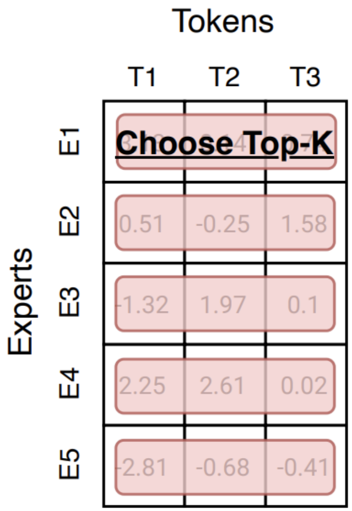
- Routing function
- Expert sizes
- Training objectives

Routing function - overview

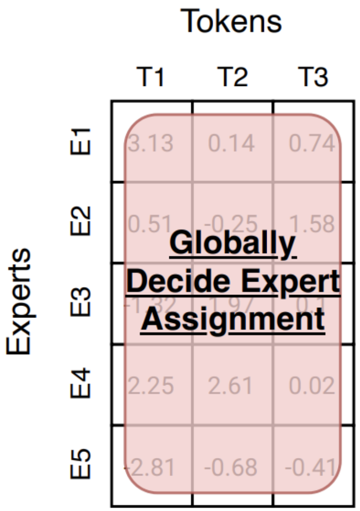
Many of the routing algorithms boil down to 'choose top k'



Token chooses expert



Expert chooses token

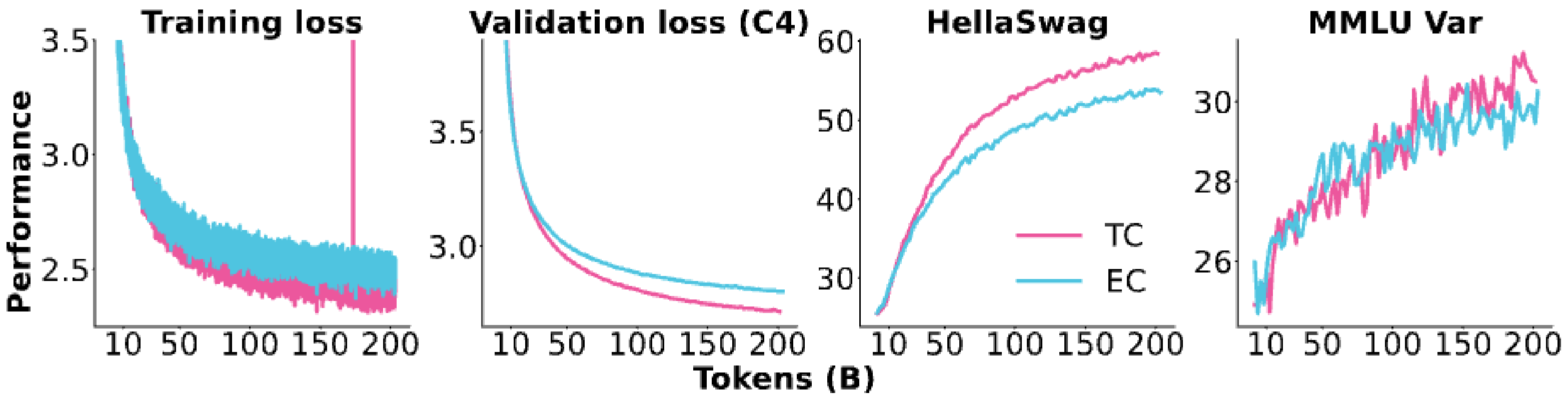


Global routing via optimization

[Fedus et al 2022]

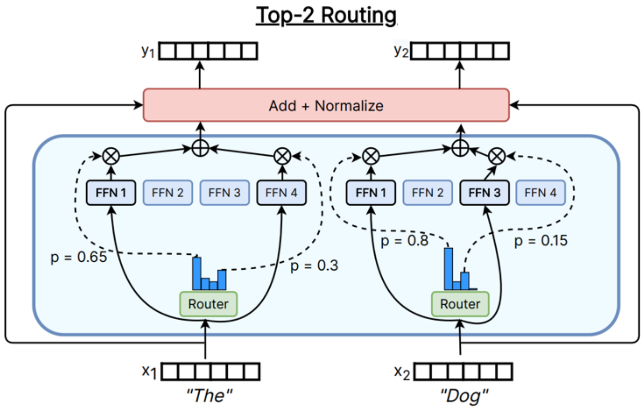
Routing type

Almost all the MoEs do a standard 'token choice topk' routing. Some recent ablations



Common routing variants in detail

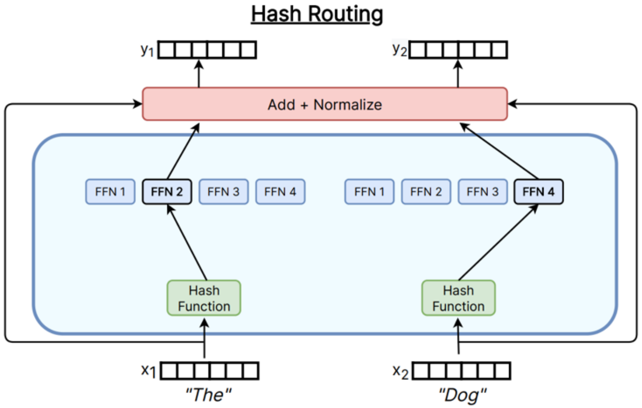
Top-k



Used in *most* MoEs

Switch Transformer (k=1)
Gshard (k=2), Grok (2), Mixtral (2), Qwen (4), DBRX (4), DeepSeek (7)

Hashing

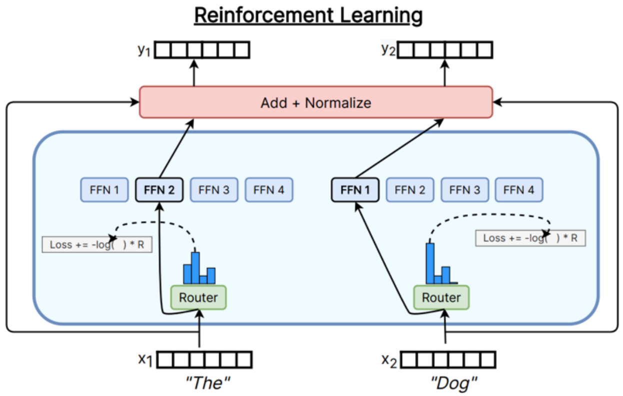


Common baseline

[Fedus et al 2022]

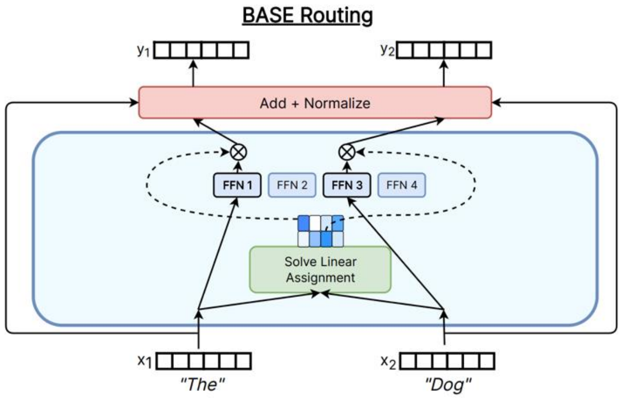
Other routing methods

RL to learn routes



Used in some of the earliest work
Bengio 2013, not common now

Solve a matching problem



Linear assignment for routing
Used in various papers like Clark ‘22

[Fedus et al 2022]

Top-K routing in detail.

Most papers do the old and classic top-k routing. How does this work?

Gating

$$\mathbf{h}_t^l = \sum_{i=1}^{N} \left(g_{i,t} \, \mathrm{FFN}_i \left(\mathbf{u}_t^l \right) \right) + \mathbf{u}_t^l,$$

$$g_{i,t} = \begin{cases} s_{i,t}, & s_{i,t} \in \mathrm{Topk}(\{s_{j,t} | 1 \leqslant j \leqslant N\}, K), \\ 0, & \text{otherwise}, \end{cases}$$

$$s_{i,t} = \mathrm{Softmax}_i \left(\mathbf{u}_t^{l^T} \mathbf{e}_i^l \right),$$

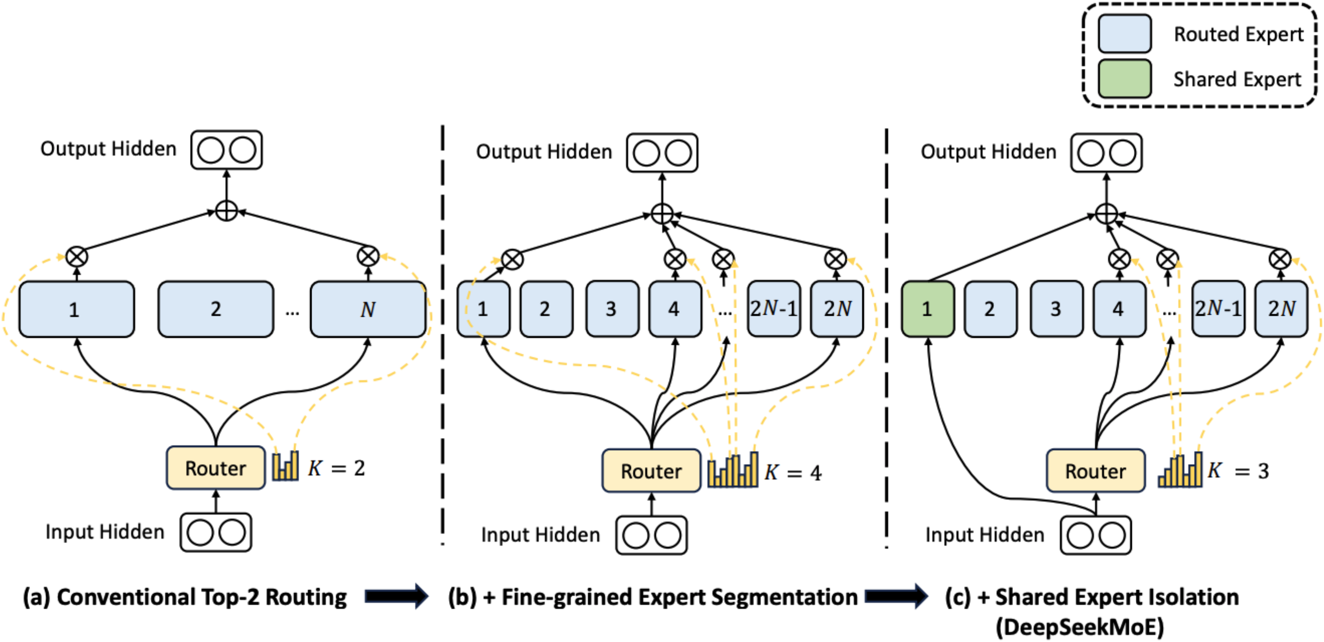
Gates selected by a logistic regressor

This is the DeepSeek (V1-2) router (Grok, Qwen do this too)

Mixtral, DBRX, DeepSeek v3 softmaxes after the TopK

[Dai et al 2024]

Recent variations from DeepSeek and other Chinese LMs



Smaller, larger number of experts + a few shared experts that are always on.

(Used in DeepSeek / Qwen, originally from DeepSpeed MoE)

Various ablations from the DeepSeek paper

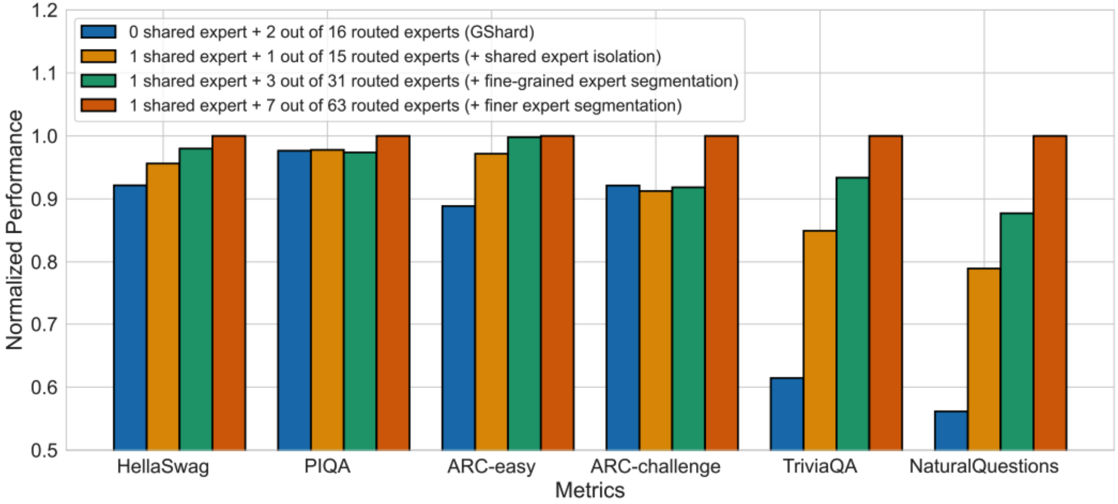
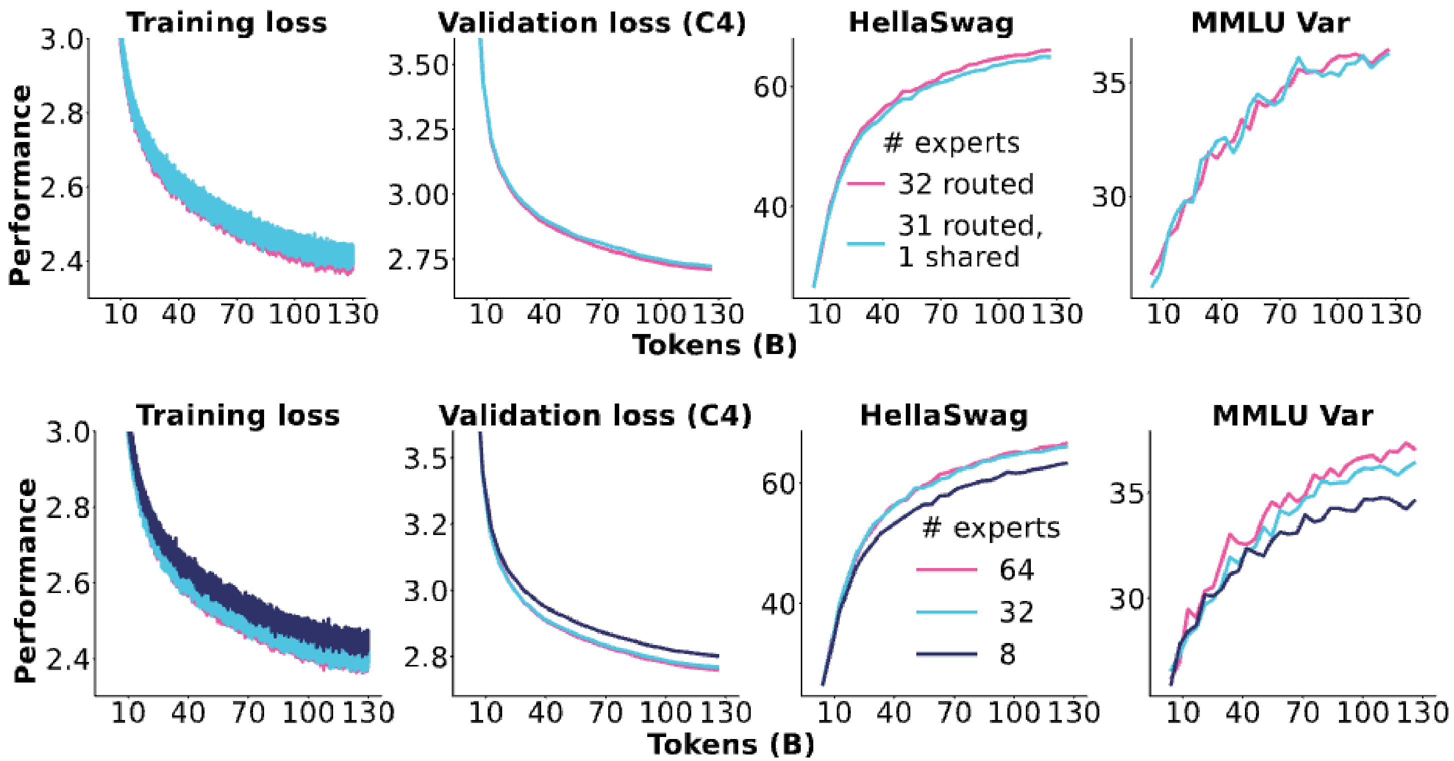


Figure 3 | Ablation studies for DeepSeekMoE. The performance is normalized by the best performance for clarity in presentation. All compared models have the same number of parameters and activated parameters. We can find that fine-grained expert segmentation and shared expert isolation both contribute to stronger overall performance.

More experts, shared experts all seem to generally help

Ablations from OlMoE

Gains from fine-grained experts, none from shared experts.



Expert routing setups for recent MoEs

Model	Routed	Active	Shared	Fine-grained ratio
GShard	2048	2	0	
Switch Transformer	64	1	0	
ST-MOE	64	2	0	
Mixtral	8	2	0	
DBRX	16	4	0	
Grok	8	2	0	
DeepSeek v1	64	6	2	1/4
Qwen 1.5	60	4	4	1/8
DeepSeek v3	256	8	1	1/14
OlMoE	64	8	0	1/8
MiniMax	32	2	0	~1/4
Llama 4 (maverick)	128	1	1	1/2

How do we train MoEs?

Major challenge: we need sparsity for training-time efficiency…

But sparse gating decisions are not differentiable!

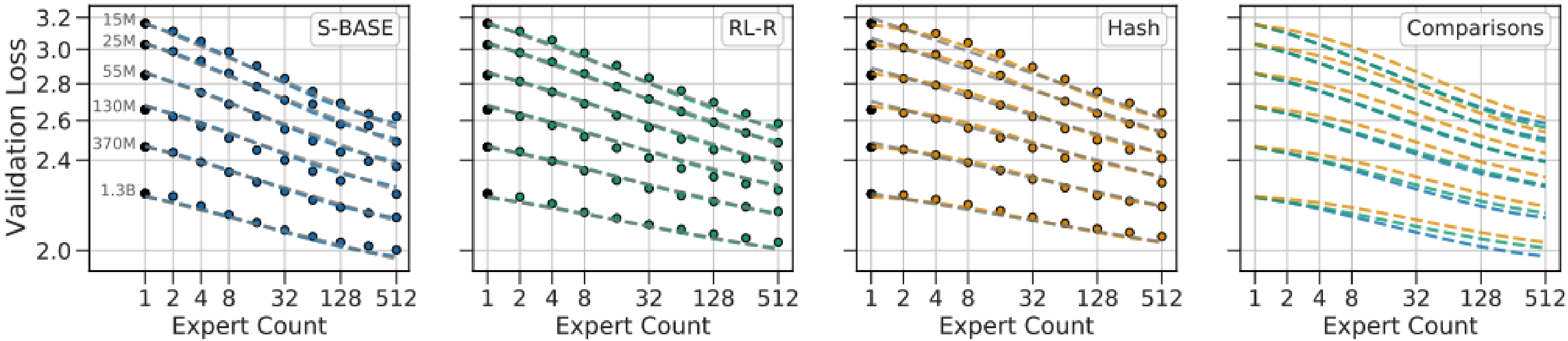
Solutions?

1. Reinforcment learning to optimize gating policies
2. Stochastic perturbations
3. Heuristic 'balancing' losses.

Guess which one people use in practice?

RL for MoEs

RL via REINFORCE does work, but not so much better that it's a clear win



(REINFORCE baseline approach, Clark et al 2020)

RL is the 'right solution' but gradient variances and complexity means it's not widely used

Stochastic approximations

$$G(x) = Softmax(KeepTopK(H(x), k))$$

$$H(x)_i = (x \cdot W_g)_i + StandardNormal() \cdot Softplus((x \cdot W_{noise})_i)$$

$$KeepTopK(v, k)_i = \begin{cases} v_i & \text{if } v_i \text{ is in the top } k \text{ elements of } v. \\ -\infty & \text{otherwise.} \end{cases}$$

From Shazeer et al 2017 – routing decisions are *stochastic* with gaussian perturbations.

1. This naturally leads to experts that are a bit more robust.
2. The softmax means that the model learns how to rank K experts

Stochastic approximations

```
if is_training:
    # Add noise for exploration across experts.
    router_logits += mtf.random_uniform(shape=router_logits.shape, minval=1-eps, maxval=1+eps)

# Convert input to softmax operation from bfloat16 to float32 for stability.
router_logits = mtf.to_float32(router_logits)

# Probabilities for each token of what expert it should be sent to.
router_probs = mtf.softmax(router_logits, axis=-1)
```

Stochastic jitter in Fedus et al 2022. This does a uniform multiplicative perturbation for the same goal of getting less brittle experts. This was later removed in Zoph et al 2022

Method	Fraction Stable	Quality (↑)
Baseline	4/6	**-1.755** $\pm$0.02
Input jitter (10^{-2})	3/3	-1.777 $\pm$0.03
Dropout (0.1)	3/3	-1.822 $\pm$0.11

Heuristic balancing losses

Another key issue – systems efficiency requires that we use experts evenly..

For each Switch layer, this auxiliary loss is added to the total model loss during training. Given N experts indexed by $i = 1$ to N and a batch $\mathcal{B}$ with T tokens, the auxiliary loss is computed as the scaled dot-product between vectors f and P,

$$\text{loss} = \alpha \cdot N \cdot \sum_{i=1}^{N} f_i \cdot P_i \tag{4}$$

where f_i is the fraction of tokens dispatched to expert i,

$$f_i = \frac{1}{T} \sum_{x \in \mathcal{B}} \mathbb{1}\{\text{argmax}\, p(x) = i\} \tag{5}$$

and P_i is the fraction of the router probability allocated for expert i, [2]

$$P_i = \frac{1}{T} \sum_{x \in \mathcal{B}} p_i(x). \tag{6}$$

From the Switch Transformer [Fedus et al 2022]

The derivative with respect to $p_i(x)$ is $\frac{\alpha N}{T^2} \sum 1_{argmax\ p(x)=i}$,
so more frequent use = stronger downweighting

Example from deepseek (v1-2)

Per-expert balancing – same as the switch transformer

$$\mathcal{L}_{\text{ExpBal}} = \alpha_1 \sum_{i=1}^{N'} f_i P_i, \tag{12}$$

$$f_i = \frac{N'}{K'T} \sum_{t=1}^{T} \mathbb{1}(\text{Token } t \text{ selects Expert } i), \tag{13}$$

$$P_i = \frac{1}{T} \sum_{t=1}^{T} s_{i,t}, \tag{14}$$

Per-device balancing – the objective above, but aggregated by device.

$$\mathcal{L}_{\text{DevBal}} = \alpha_2 \sum_{i=1}^{D} f_i' P_i', \tag{15}$$

$$f_i' = \frac{1}{|\mathcal{E}_i|} \sum_{j \in \mathcal{E}_i} f_j, \tag{16}$$

$$P_i' = \sum_{j \in \mathcal{E}_i} P_j, \tag{17}$$

DeepSeek v3 variation – per-expert biases

Set up a per-expert bias (making it more likely to get tokens) and use online learning

$$g'_{i,t} = \begin{cases} s_{i,t}, & s_{i,t} + b_i \in \operatorname{Topk}(\{s_{j,t} + b_j | 1 \leqslant j \leqslant N_r\}, K_r), \\ 0, & \text{otherwise.} \end{cases}$$

They call this '**auxiliary loss free balancing**'

Complementary Sequence-Wise Auxiliary Loss. Although DeepSeek-V3 mainly relies on the auxiliary-loss-free strategy for load balance, to prevent extreme imbalance within any single sequence, we also employ a complementary sequence-wise balance loss:

$$\mathcal{L}_{\mathrm{Bal}} = \alpha \sum_{i=1}^{N_r} f_i P_i, \tag{17}$$

$$f_i = \frac{N_r}{K_r T} \sum_{t=1}^{T} \mathbb{1}\left(s_{i,t} \in \operatorname{Topk}(\{s_{j,t} | 1 \leqslant j \leqslant N_r\}, K_r)\right), \tag{18}$$

$$s'_{i,t} = \frac{s_{i,t}}{\sum_{j=1}^{N_r} s_{j,t}}, \tag{19}$$

$$P_i = \frac{1}{T} \sum_{t=1}^{T} s'_{i,t}, \tag{20}$$

(but the approach is not fully aux loss free..)

What happens when removing load balancing losses?

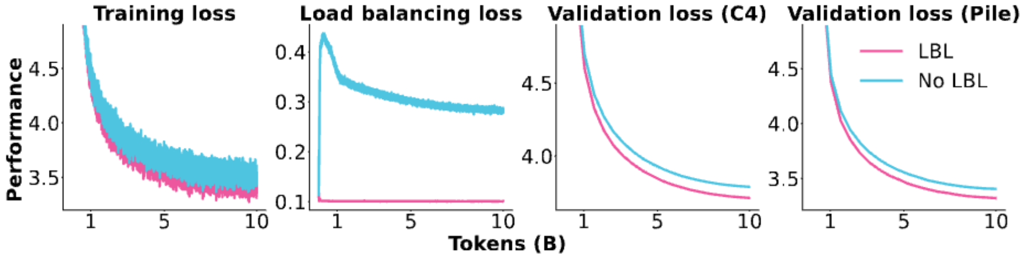


Figure 9: **Impact of applying a load balancing loss (LBL).** The training loss plot excludes the load balancing loss for both models. More results, logs, and configurations: https://wandb.ai/ai2-llm/olmoe/reports/Plot-LBL-vs-No-LBL--Vmlldzo4OTkyNDg4

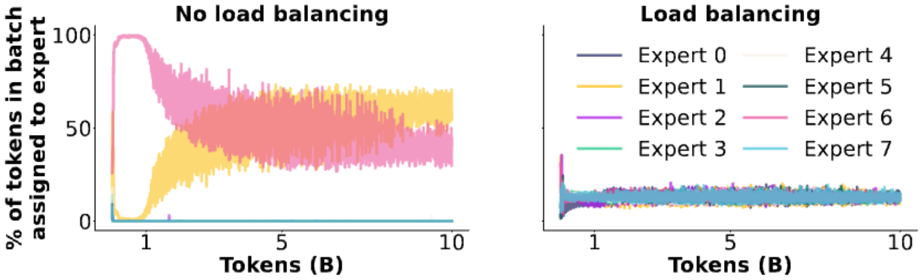
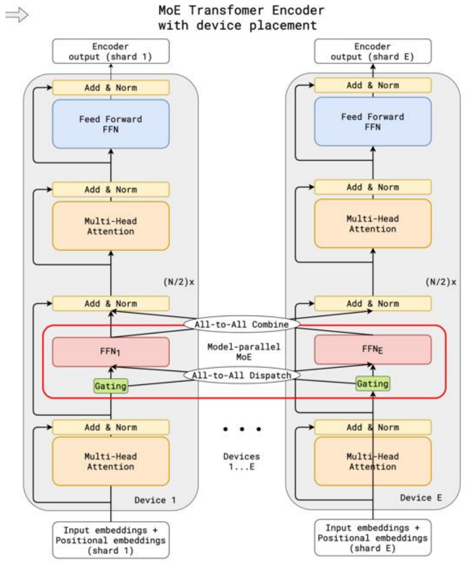


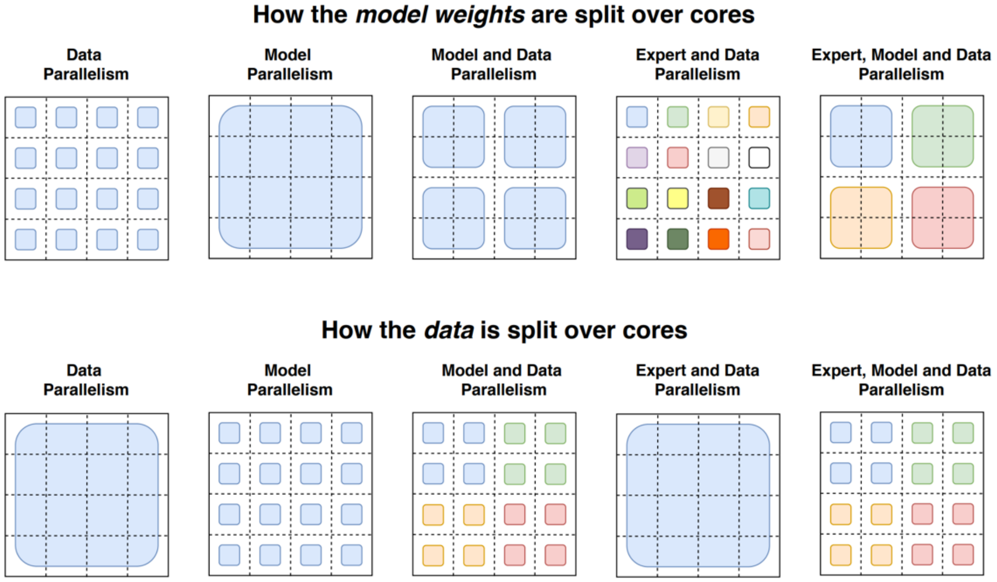
Figure 10: **Expert assignment during training when using or not using a load balancing loss for the first MoE layer.** More results, logs, and configurations: https://wandb.ai/ai2-llm/olmoe/reports/Plot-LBL-vs-No-LBL--Vmlldzo4OTkyNDg4

Training MoEs – the systems side

MoEs parallelize nicely – Each FFN can fit in a device

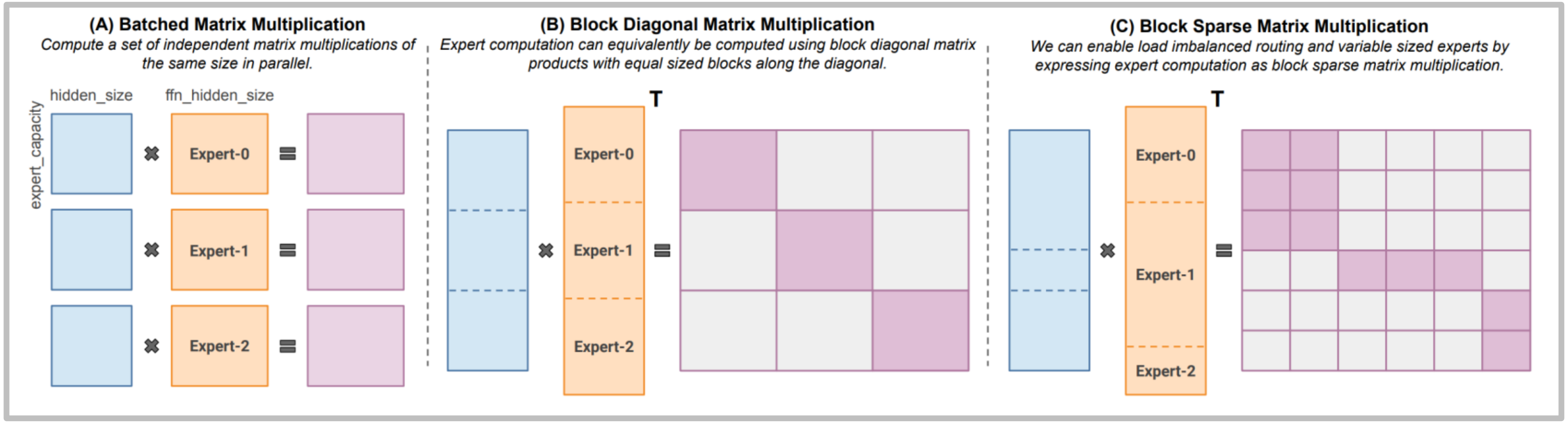


Enables additional kinds of parallelism



Training MoEs – the systems side

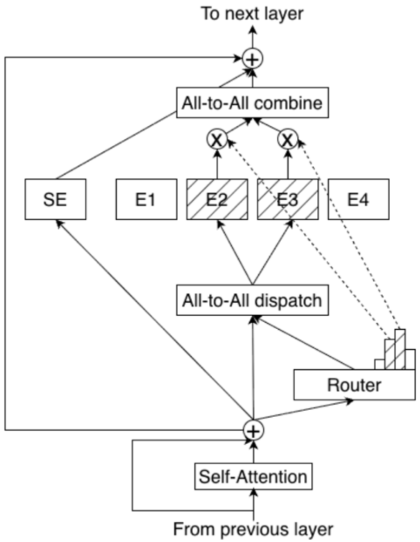
MoE routing allows for parallelism, but also some complexities



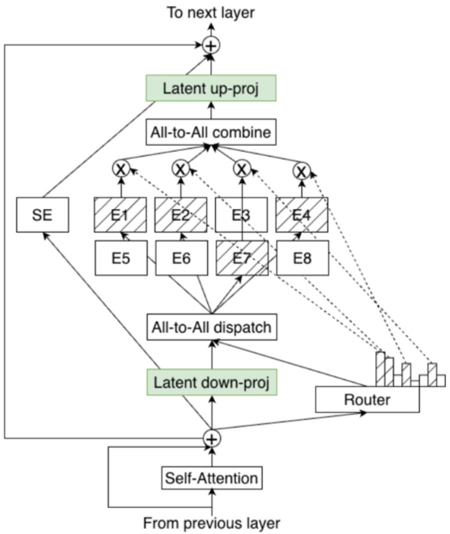
Modern libraries like MegaBlocks (used in many open MoEs) use smarter sparse MMs

MoE parallelism and architecture modifications

2.2. LatentMoE: Hardware-Aware Expert Design for Improved Accuracy per Byte



(a) Standard MoE architecture.



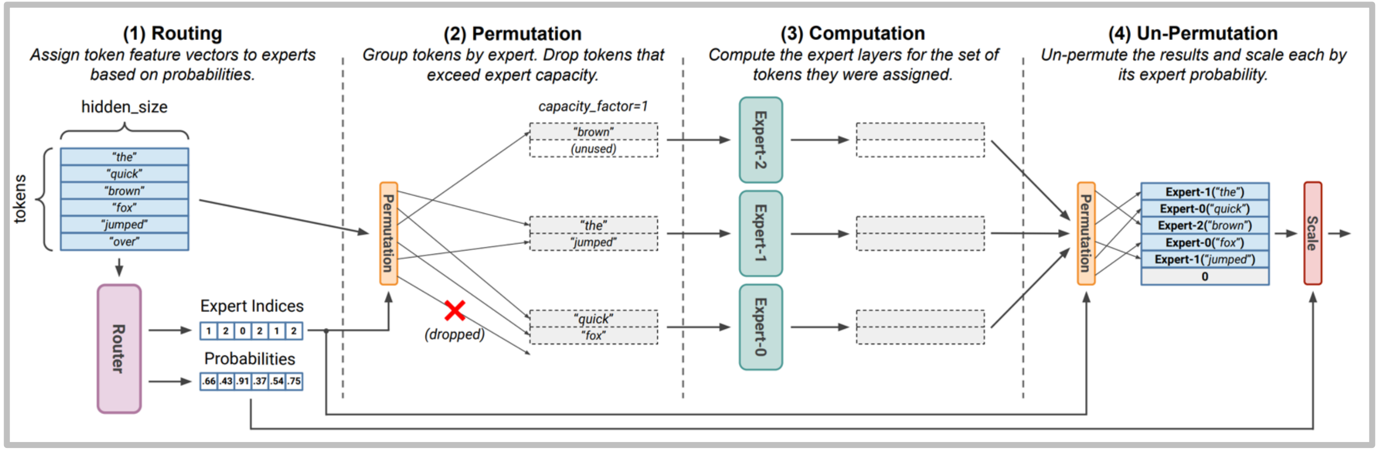
(b) LatentMoE architecture.

New ideas from Nemotron 3 – down-projecting the activations to reduce communication

Fun side issue – stochasticity of MoE models

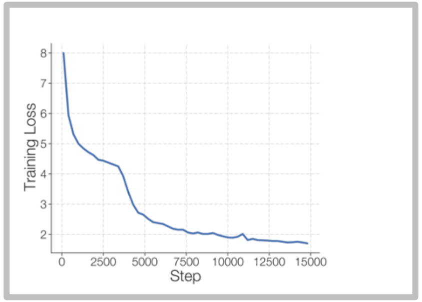
MoEs can have additional stochasticity beyond normal models..

Why would a MoE have additional randomness?



Token dropping from routing happens at a *batch* level – this means that other people's queries can drop your token!

Issues with MoEs - stability



[7] Exponential functions have the property that a small input perturbation can lead to a large difference in the output. As an example, consider inputting 10 logits to a softmax function with values of 128 and one logit with a value 128.5. A roundoff error of 0.5 in `bfloat16` will alter the softmax output by 36% and incorrectly make all logits equal. The calculation goes from $\frac{\exp(0)}{\exp(0)+10\cdot\exp(-0.5)} \approx 0.142$ to $\frac{\exp(0)}{\exp(0)+10\cdot\exp(0)} \approx 0.091$. This occurs because the max is subtracted from all logits (for numerical stability) in softmax operations and the roundoff error changes the number from 128.5 to 128. This example was in `bfloat16`, but analogous situations occur in `float32` with larger logit values.

[Zoph 2022]

Solution: Use Float 32 just for the expert router (sometimes with an aux z-loss)

$$L_z(x) = \frac{1}{B}\sum_{i=1}^{B}\left(\log\sum_{j=1}^{N} e^{x_j^{(i)}}\right)^2 \tag{5}$$

Z-loss stability for the router

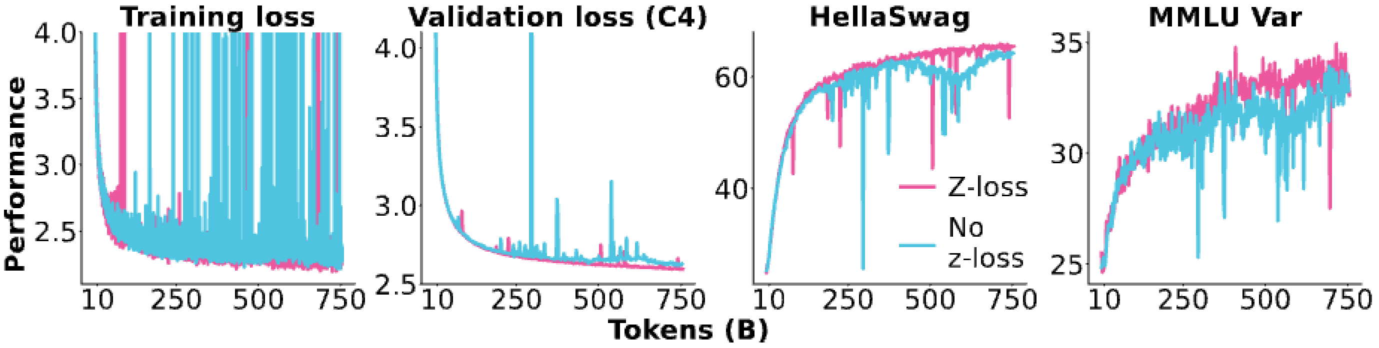
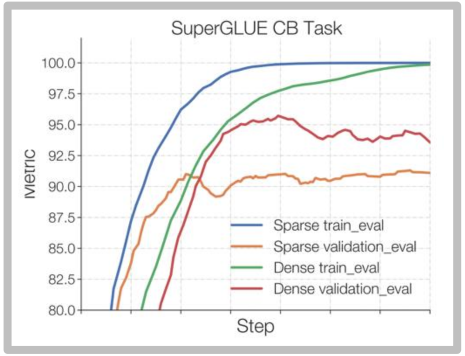


Figure 11: **Router z-loss.** We compare adding router z-loss with a loss weight of 0.001 versus no additional z-loss. More results, logs, and configurations: https://wandb.ai/ai2-llm/olmoe/reports/Plot-Zloss-vs-none--Vmlldzo4NDM4NjUz

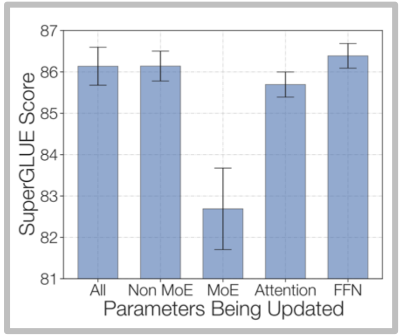
What happens when we remove the z-loss?

Issues with MoEs – fine-tuning

Sparse MoEs can overfit on smaller fine-tuning data



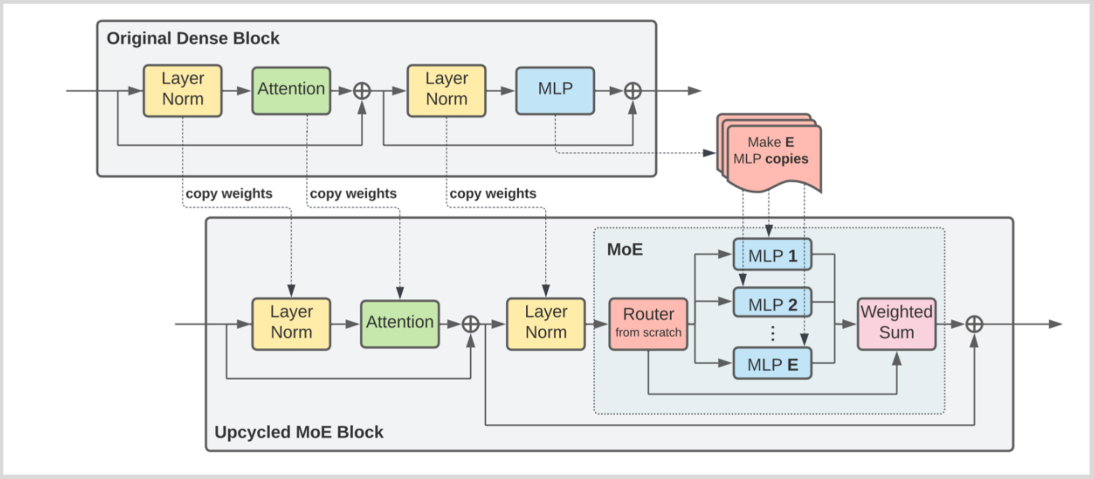
Zoph et al solution – finetune non-MoE MLPs

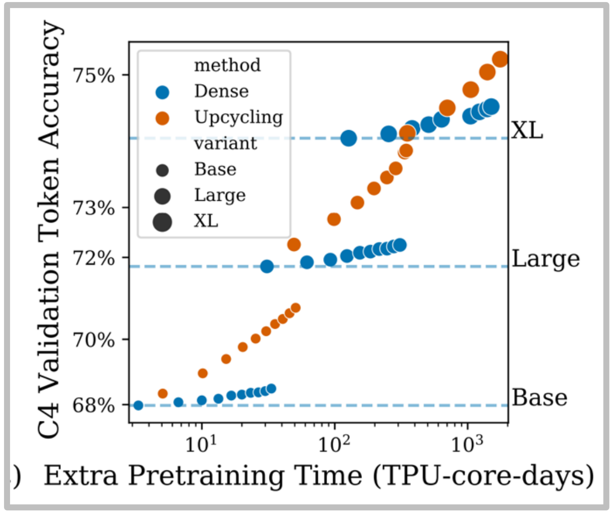


DeepSeek solution – use lots of data 1.4M SFT

Training Data. For training the chat model, we conduct supervised fine-tuning (SFT) on our in-house curated data, comprising 1.4M training examples. This dataset spans a broad range of categories including math, code, writing, question answering, reasoning, summarization, and more. The majority of our SFT training data is in English and Chinese, rendering the chat model versatile and applicable in bilingual scenarios.

Other training methods - upcycling





Can we use a pre-trained LM to initialize a MoE?

Upcycling example - MiniCPM

Uses the MiniCPM model (topk=2, 8 experts, ~ 4B active params).

Model	C-Eval	CMMLU	MMLU	HumanEval	MBPP	GSM8K	MATH	BBH
Llama2-34B	-	-	62.6	22.6	33.0†	42.2	6.24	**44.1**
Deepseek-MoE (16B)	40.6	42.5	45.0	26.8	39.2	18.8	4.3	-
Mistral-7B	46.12	42.96	**62.69**	27.44	45.20	33.13	5.0	41.06
Gemma-7B	42.57	44.20	60.83	38.41	50.12	47.31	6.18	39.19
MiniCPM-2.4B	51.13	51.07	53.46	50.00	47.31	53.83	10.24	36.87
MiniCPM-MoE (13.6B)	**58.11**	**58.80**	58.90	**56.71**	**51.05**	**61.56**	**10.52**	39.22

Table 6: Benchmark results of MiniCPM-MoE. † means evaluation results on the full set of MBPP, instead of the hand-verified set (Austin et al., 2021). The evaluation results of Llama2-34B and Qwen1.5-7B are taken from their technical reports.

Simple MoE, shows gains from the base model with ~ 520B tokens for training

Upcycling example – Qwen MoE

Qwen MoE – Initialized from the Qwen 1.8B model top-k=4, 60 experts w/ 4 shared.

Model	#Parameters	#(Activated) Parameters	MMLU	GSM8K	HumanEval	Multilingual	MT-Bench
Mistral-7B	7.2	7.2	64.1	47.5	27.4	40.0	7.60
Qwen1.5-7B	7.7	7.7	64.6	50.9	32.3	-	-
Gemma-7B	8.5	7.8	61.0	62.5	36.0	45.2	7.60
DeepSeekMoE 16B	16.4	2.8	45.0	18.8	26.8	-	6.93
Qwen1.5-MoE-A2.7B	14.3	2.7	62.5	61.5	34.2	40.8	7.17

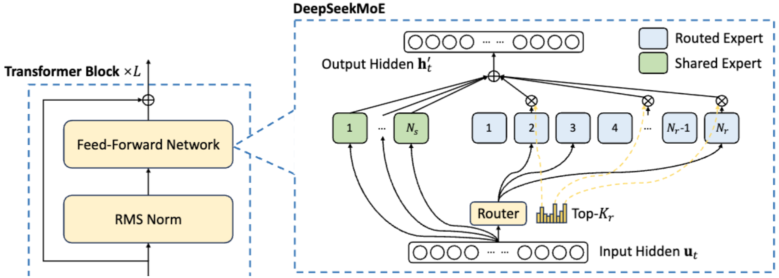
Similar architecture / setup to DeepSeekMoE, but one of the first (confirmed) upcycling successes

DeepSeek MoE v1-v2-v3

To wrap up, we'll walk through the DeepSeek MoE architecture.

V1 (16B – 2.8 active):

Shared (2) + Fine-grained (64/4) experts



Standard, top-k routing

$$\mathbf{h}_t' = \mathbf{u}_t + \sum_{i=1}^{N_s} \mathrm{FFN}_i^{(s)}(\mathbf{u}_t) + \sum_{i=1}^{N_r} g_{i,t}\, \mathrm{FFN}_i^{(r)}(\mathbf{u}_t),$$

$$g_{i,t} = \begin{cases} s_{i,t}, & s_{i,t} \in \mathrm{Topk}(\{s_{j,t} | 1 \leqslant j \leqslant N_r\}, K_r), \\ 0, & \text{otherwise}, \end{cases}$$

$$s_{i,t} = \mathrm{Softmax}_i\left({\mathbf{u}_t}^T \mathbf{e}_i\right),$$

Standard Aux-loss balancing (Expert + Device)

$$\mathcal{L}_{\mathrm{ExpBal}} = \alpha_1 \sum_{i=1}^{N_r} f_i P_i,$$

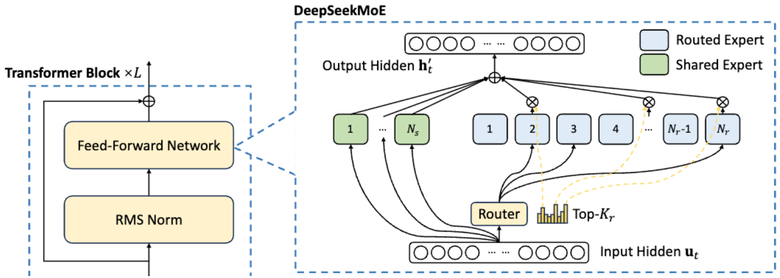
$$f_i = \frac{N_r}{K_r T} \sum_{t=1}^{T} \mathbb{1}(\text{Token } t \text{ selects Expert } i),$$

$$P_i = \frac{1}{T} \sum_{t=1}^{T} s_{i,t},$$

DeepSeek MoE v2

V2 (236B – 21 active):

Shared (2) + Fine-grained (160/10) experts, 6 active



New things:

Top-M device routing

For DeepSeek-V2, beyond the naive top-K selection of routed experts, we additionally ensure that the target experts of each token will be distributed on at most M devices. To be specific, for each token, we first select M devices that have experts with the highest affinity scores in them. Then, we perform top-K selection among experts on these M devices. In practice, we find that when $M \geqslant 3$, the device-limited routing can achieve a good performance roughly aligned with the unrestricted top-K routing.

Communication balancing loss – balancing both communication in *and* out

$$\mathcal{L}_{\mathrm{CommBal}} = \alpha_3 \sum_{i=1}^{D} f_i'' P_i'', \tag{29}$$

$$f_i'' = \frac{D}{MT} \sum_{t=1}^{T} \mathbb{1}(\text{Token } t \text{ is sent to Device } i), \tag{30}$$

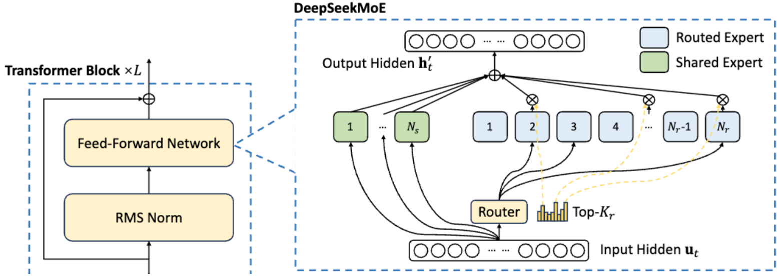
$$P_i'' = \sum_{j \in \mathcal{E}_i} P_j, \tag{31}$$

where α_3 is a hyper-parameter called communication balance factor. The device-limited routing mechanism operates on the principle of ensuring that each device transmits at most MT hidden states to other devices. Simultaneously, the communication balance loss is employed to encourage each device to receive around MT hidden states from other devices. The communication balance loss guarantees a balanced exchange of information among devices, promoting efficient communications.

DeepSeek MoE v3

V2 (671B – 37 active):

Shared (1) + Fine-grained (258) experts, 8 active



New things

Sigmoid+Softmax topK + topM

$$\mathbf{h}_t' = \mathbf{u}_t + \sum_{i=1}^{N_s} \mathrm{FFN}_i^{(s)}(\mathbf{u}_t) + \sum_{i=1}^{N_r} g_{i,t}\, \mathrm{FFN}_i^{(r)}(\mathbf{u}_t),$$

$$g_{i,t} = \frac{g_{i,t}'}{\sum_{j=1}^{N_r} g_{j,t}'},$$

$$g_{i,t}' = \begin{cases} s_{i,t}, & s_{i,t} \in \mathrm{Topk}(\{s_{j,t} | 1 \leqslant j \leqslant N_r\}, K_r), \\ 0, & \text{otherwise,} \end{cases}$$

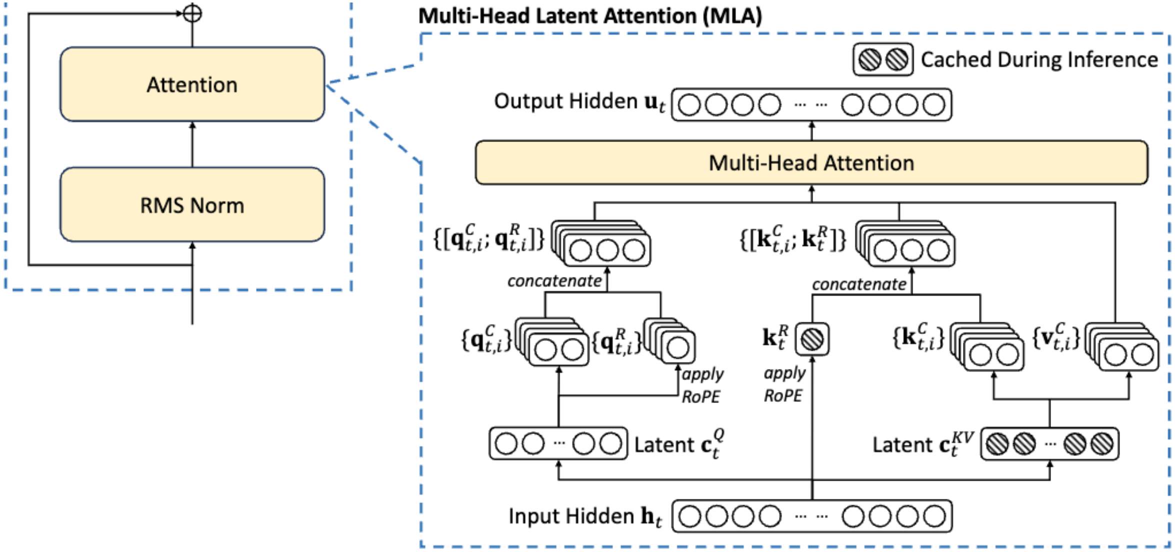
$$s_{i,t} = \mathrm{Sigmoid}\left(\mathbf{u}_t{}^T \mathbf{e}_i\right),$$

Aux-loss-free + seq-wise aux

$$g_{i,t}' = \begin{cases} s_{i,t}, & s_{i,t} + b_i \in \mathrm{Topk}(\{s_{j,t} + b_j | 1 \leqslant j \leqslant N_r\}, K_r), \\ 0, & \text{otherwise.} \end{cases}$$

Bonus: What else do you need to make DeepSeek MoE v3?

MLA : Multihead, latent attention

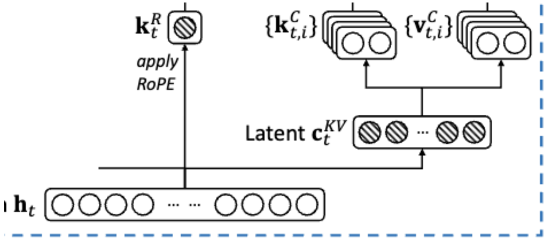


Basic idea: express the Q, K, V as functions of a lower-dim, 'latent' activation

What else do you need to make DeepSeek MoE v3?

Basic idea: express the Q, K, V as functions of a lower-dim, 'latent' activation

$$\mathbf{c}_t^{KV} = W^{DKV}\mathbf{h}_t,$$
$$\mathbf{k}_t^{C} = W^{UK}\mathbf{c}_t^{KV},$$
$$\mathbf{v}_t^{C} = W^{UV}\mathbf{c}_t^{KV},$$



Benefits: when KV-caching, we only need to store c_t^{KV}, which can be much smaller.
W^{UK} can be merged into the Q projection

(they also compress queries, for memory savings during training)

$$\mathbf{c}_t^{Q} = W^{DQ}\mathbf{h}_t,$$
$$\mathbf{q}_t^{C} = W^{UQ}\mathbf{c}_t^{Q},$$

Complexity: rope conflicts with MLA-style caching.

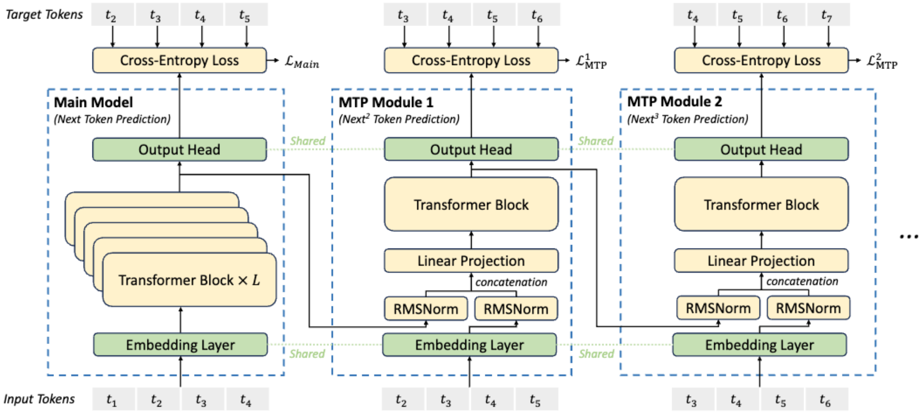
Without RoPE – $\langle Q, K\rangle = \left\langle hW^Q, W^{UK}c_t^{KV}\right\rangle = \langle h\, W^Q W^{UK}, c_t^{KV}\rangle$

With RoPE - $\langle QR_q, R_k K\rangle = \left\langle hW^Q R_q, R_k W^{UK} c_t^{KV}\right\rangle = \left\langle h\, W^Q R_q R_k W^{UK}, c_t^{KV}\right\rangle$

The solution – Have a few non-latent key dimensions that can be rotated

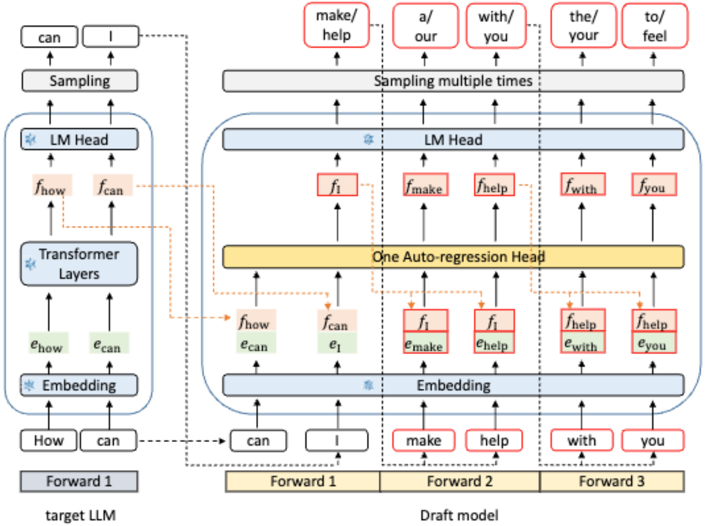
What else do you need to make DeepSeek MoE v3?

MTP: Have small, lightweight models that predict multiple steps ahead



(But they only do MTP with one token ahead)

[Deepseek v3]



[EAGLE]

$$\mathbf{h}_i'^k = M_k[\mathrm{RMSNorm}(\mathbf{h}_i^{k-1}); \mathrm{RMSNorm}(\mathrm{Emb}(t_{i+k}))],$$

$$\mathbf{h}_{1:T-k}^k = \mathrm{TRM}_k(\mathbf{h}_{1:T-k}'^k),$$

$$P_{i+k+1}^k = \mathrm{OutHead}(\mathbf{h}_i^k).$$

(See paper for ablations)

MoE summary

- MoEs take advantage of sparsity – not all inputs need the full model
- Discrete routing is hard, but top-k heuristics seem to work
- Lots of empirical evidence now that MoEs work, and are cost-effective